
Build Deployment Maven Build Execution:

Purpose of Build Deployment Phase

This phase is used to convert your automation project into a proper executable build and verify that the entire framework is working correctly using Maven commands.

In simple words

This phase ensures your Selenium + TestNG + CSV framework is ready to run as a complete project without manual execution of individual files.

What is Maven Build Execution

Maven is a build management tool used in Java projects. In your SauceDemo automation framework, Maven is used to

- manage dependencies
- compile project code
- execute test cases
- generate reports
- create build structure

Why Build Deployment is Important

It is important because

- ensures all dependencies are correctly downloaded
- confirms project structure is correct
- validates testng.xml execution
- checks if automation framework is ready for CI CD integration
- helps in generating final build output

Maven Build Lifecycle Used in Your Project

In your framework, mainly these phases are used

- compile
- test
- package

Step by Step Build Execution Flow

Step 1 Clean Project

Removes previous build files

Example command

```
mvn clean
```

Step 2 Compile Project

Compiles all Java files

Example command

```
mvn compile
```

This ensures

BaseClass

LoginPage

Utilities

Test classes

are error free

Step 3 Execute Test Cases

Runs TestNG test suite

Example command

```
mvn test
```

This will

execute LoginTest.java

read data from CSV file

run multiple login scenarios

generate logs and screenshots

Step 4 Generate Reports

After execution Maven helps generate

Extent Report

TestNG Report

These reports are stored in Reports folder

Example of Maven Execution Output

Here is a real time flow for your SauceDemo project

Starting Maven Build Execution
Compiling project source code
Dependencies downloaded successfully

TestNG suite execution started
Launching Chrome browser
Opening SauceDemo application

Executing TC_001 with standard_user
Login successful
Redirected to Products page

Executing TC_002 with invalid_user
Login failed
Error message displayed

Test execution completed
Generating Extent Report
Build SUCCESS

What Happens After Build Execution

After successful build

Reports folder is updated
Execution.log is generated
Screenshots are stored in Fail or Pass folder
TestNG and Extent reports are created
Defect logs can be reviewed

Real Project Connection

In your framework structure

pom.xml controls build
testng.xml controls execution

BaseClass handles browser setup
CSVReader provides data
ExtentManager generates report

All these are executed together during Maven build
